# Opgaver kap. 14

14-56 56. A wire is under 32.8-N tension, carrying a wave described by $y = 1.75 \sin(0.211x - 466t)$, where $x$ and $y$ are in centimeters and $t$ is in seconds. What are (a) the wave amplitude, (b) the wavelength, (c) the wave period, (d) the wave speed, and (e) the power carried by the wave?

14.59 57. At a point 12 m from a source of spherical sound waves, you measure the intensity 690 mW/m$^2$. How far do you need to walk, directly away from the source, until the intensity is 260 mW/m$^2$?

14.67 65. At 2.0 m from a localized sound source you measure the intensity level as 75 dB. How far away must you be for the perceived loudness to drop in half (i.e., to an intensity level of 65 dB)?

14.79 77. You're a meteorologist specifying a new Doppler radar system that determines the velocity of distant raindrops by reflecting radar signals (which travel at the speed of light) off them and measuring the Doppler shift. You need a system that will measure speeds as low as 2.5 km/h. A vendor offers a 5.0-GHz radar that can detect a frequency shift of only 50 Hz. Is that sufficient?

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